

markdown

TECHNICAL SPECIFICATION BLUEPRINT: STATE-OF-ART ORPHAN SCENE DETECTOR

Document Reference: AR-OSD-2026-SPEC

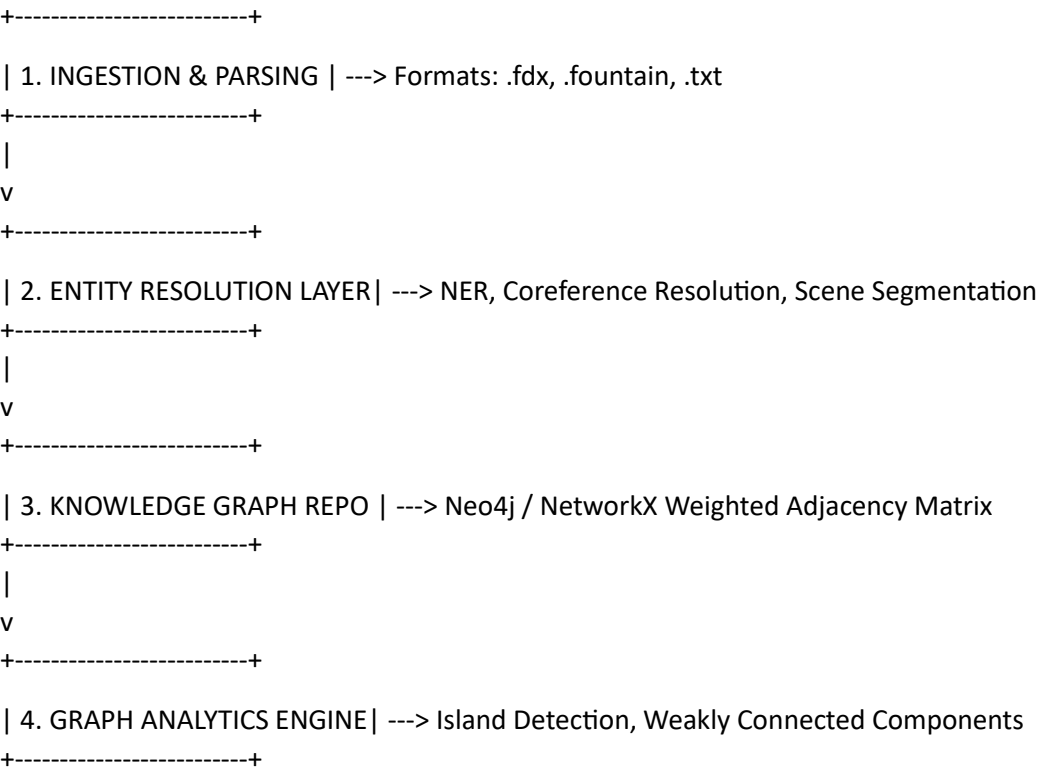
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Classification: Technical Architecture Specification

1. SYSTEM ARCHITECTURE OVERVIEW

The State-of-the-Art Orphan Scene Detector (OSD) treats a screenplay as a multi-layered, semantic Network Graph (Graph Database / Directed Acyclic Graph structure). The core architecture is divided into four strictly isolated pipeline layers to ensure scalability, precision, and deterministic testing.

Use code with caution.



2. DATA INGESTION & STRUCTURAL PARSING

The ingestion engine must parse raw screenwriting file formats into a structured Object Model.

2.1 Supported Document Specs

* ****Final Draft (.fdx):**** XML-based parsing mapping explicit element tags (<Key> structural tags for Scene Headings, Action, Character, Dialogue, Parenthetical).

* ****Fountain / Plain Text:**** Regex-driven state machine tracking indentations, uppercase markers, and dual-dialogue breaks.

2.2 The Scene Object Model (JSON Schema)

Every isolated scene must be instantiated into the following data structure:

```
``json
{
  "scene_id": "int (1-based index)",
  "scene_heading": {
    "type": "INT / EXT / INT-EXT",
    "location": "string (normalized)",
    "time_of_day": "string (DAY / NIGHT / LATER / CONTINUOUS)"
  },
  "raw_text": "string",
  "entities": {
    "characters_speaking": ["string"],
    "characters_mentioned": ["string"],
    "props_detected": ["string"],
    "wardrobe_detected": ["string"]
  },
  "embeddings": {
    "dialogue_vector": "float[384]",
    "action_vector": "float[384]"
  }
}
```

}

...

3. MULTI-LAYERED LINKAGE MATRIX (THE EDGES)

To achieve a foolproof mechanism, an "edge" (connection) between Scene S_i and Scene S_j is determined by a strict, deterministic multi-variable mathematical scoring matrix.

$$\text{Total Link Weight } (W_{ij}) = \alpha \cdot C_{ij} + \beta \cdot L_{ij} + \gamma \cdot P_{ij} + \delta \cdot E_{ij}$$

Where:

- * $\alpha = 0.40$ (Character Continuity Weight)
- * $\beta = 0.25$ (Spatial/Location Continuity Weight)
- * $\gamma = 0.20$ (Prop / Narrative Tracking Weight)
- * $\delta = 0.15$ (Semantic / LLM Dialogical Overlap Weight)

3.1 Edge Validation Rules

1. **Character Linkage (C_{ij}):** Calculated using Jaccard Similarity of the union of `characters\_speaking` and `characters\_mentioned` between both scenes.

* **Coreference Resolution Constraint:** If "John" is referenced as "he" or "the detective" in adjacent action lines, an NLP coreference model (e.g., SpaCy Coref) must map these to the target entity ID "JOHN".

2. **Spatial Linkage (L_{ij}):** Binary flag (1 or 0) or Levenshtein Distance matrix matching normalized locations (e.g., "JOHN'S APARTMENT - BEDROOM" matches "JOHN'S APARTMENT").

* **Temporal Override:** If S_j features `LATER`, `SAME`, or `CONTINUOUS` in the time column, L_{ij} immediately defaults to 1 relative to S_{i-1} .

3. **Prop/Asset Linkage (P_{ij}):** String matching of items extracted via Part-of-Speech (POS) noun phrase extraction filter.

4. **Semantic Linkage (E_{ij}):** Cosine Similarity metric evaluated over text embeddings generated by a sentence-transformer model (e.g., `all-MiniLM-L6-v2`).

4. FOOLPROOF ORPHAN DETECTION LOGIC

A scene is mathematically classified as an **Orphan** or a **Dangling Node** based on topological isolation in the generated graph network.

4.1 Algorithm 1: Strict Isolation (Degree Centrality)

A node S_x is flagged as a **Hard Orphan** if its degree centrality equals zero.

$$\text{Degree}(S_x) = 0$$

4.2 Algorithm 2: Subplot Isolation (Weakly Connected Components)

Often, an entire sequence of 2 to 3 scenes connects perfectly *to each other*, but has zero connection to the rest of the 100-scene script.

* **Mechanism:** Execute Tarjan's or Kosaraju's algorithm for **Weakly Connected Components (WCC)**.

* **Flag Condition:** Any isolated subgraph component G' whose size $|G'| \leq 3$ scenes while the total graph $|G| \geq 40$ scenes is flagged as an **Orphan Subplot Chain**.

4.3 Essential Edge Cases & False-Positive Overrides

To prevent false alerts on common cinematic narrative styles, the following system exceptions must be coded into the logic layer:

| Edge Case Scene Style | Risk Profile | Graph Pattern | Programmatic Override / Safe Code |

| :--- | :--- | :--- | :--- |

| **Cold Open / Prologue** | High False Positive | Connects only to Scene 2; 0 links to body. | Exempt Scene 1 & 2 if structural tag `PROLOGUE` or page count < 3 . |

| **Flashbacks** | High False Positive | Location changes completely; props vanish. | Scan for universal character footprint. If $SC_{ij} \geq 1$ anywhere in the main cluster, override orphan flag. |

| **Intercut Phone Calls** | High False Positive | Rapid structural switching of locations. | Detect keyword `INTERCUT` in Scene Heading. Automatically group all matching sub-nodes into a single Meta-Node block before processing. |

| **Montages** | Critical False Positive | Series of micro-locations (e.g., driving through cities). |
Look for sequence of scenes containing string `MONSTAGE` or `SERIES OF SHOTS`. Treat the entire
sequence block as an individual unified node. |

5. TECHNICAL STACK RECOMMENDATIONS

For an enterprise, state-of-the-art deployment, implement the following tech stack:

- * **Parsing Layer:** Python (`python-docx` / custom regex parser) or Node.js.
- * **NLP & Feature Extraction:** `SpaCy` (for POS tagging, Entity Extraction) + `HuggingFace Transformers` (Sentence-BERT for semantic text embeddings).
- * **Graph Processing Engine:** `NetworkX` (Python-native prototyping) or `Neo4j` (Production deployment utilizing Cypher queries for Graph Clustering).
- * **Visualization Layer:** `D3.js` or `Vis.js` to render an interactive web con